

Chapter 10: Tangible Non-current Assets and Depreciation

Excel file for this chapter: [excel/Ch10_Depreciation.xlsx](#)

1. Learning goals

1. Tell the difference between non-current assets and current assets.
2. Tell the difference between capital expenditure and revenue expenditure.
3. Explain what depreciation is and why it is charged.
4. Calculate depreciation using the straight line, reducing balance and revaluation methods.
5. Record depreciation in the ledger and in the financial statements.
6. Record the sale (disposal) of a non-current asset and find the profit or loss.

2. Key words

Term	Meaning
Non-current asset	An asset bought to use in the business for more than one year , not to resell. Example: premises, machinery, vehicles.
Tangible	You can touch it. Example: a van. (Intangible: goodwill, a brand.)
Cost	Purchase price plus all costs to get the asset ready for use: delivery, installation, legal fees, first registration.
Depreciation	The part of the cost of a non-current asset used up in one year. It is an expense .
Residual value	The amount you expect to sell the asset for at the end of its useful life. Also called scrap value.
Useful life	The number of years the business expects to use the asset.
Provision for depreciation	Also called accumulated depreciation. The total depreciation charged on an asset since it was bought.
Net book value (NBV)	Cost minus accumulated depreciation. Also called carrying value.
Disposal	Selling or scrapping a non-current asset.

3. Capital and revenue expenditure

Capital expenditure	Revenue expenditure
Buying or improving a non-current asset	Day to day running costs
Benefit lasts more than one year	Benefit used up within the year
Shown in the Statement of Financial Position	Shown as an expense in the Statement of Profit or Loss
Examples: buying a van, delivery and installation of a machine, building an extension	Examples: fuel, repairs, insurance, wages, road tax

Why it matters: if you record capital expenditure as an expense, profit is **understated** and assets are **understated**. If you record revenue expenditure as an asset, profit and assets are **overstated**.

Capital **receipts** (for example, money from selling an old van, or a loan) are not revenue. Revenue **receipts** (sales, rent received, commission) go to the Statement of Profit or Loss.

4. Why do assets lose value?

Four causes of depreciation:

1. **Wear and tear:** the asset is used and wears out (a delivery van).
2. **Passage of time:** a lease runs out after a fixed number of years.
3. **Obsolescence:** the asset becomes out of date because newer models exist (old computers).
4. **Depletion:** the asset is used up (a mine or quarry).

5. Why charge depreciation?

- **Accruals (matching) concept:** the cost of the asset is spread over the years it helps earn revenue.
- **Prudence:** assets are not overstated in the Statement of Financial Position.
- It gives a true figure for profit each year.

Remember: depreciation is **not** cash. No money leaves the bank when you charge depreciation.

6. Method 1: Straight line

The same amount is charged every year.

Formula:

$$\text{Annual depreciation} = (\text{Cost} - \text{Residual value}) / \text{Useful life in years}$$

Or a fixed percentage of **cost** each year.

Worked example 10.1

A machine costs \$20,000. Expected residual value is \$2,000 after 4 years.

$$\text{Annual depreciation} = (20,000 - 2,000) / 4 = \text{\$4,500 per year}$$

Year	Cost	Depreciation for year	Accumulated depreciation	NBV at end
1	20,000	4,500	4,500	15,500
2	20,000	4,500	9,000	11,000
3	20,000	4,500	13,500	6,500
4	20,000	4,500	18,000	2,000

Best for: assets which give the same benefit each year, for example premises or furniture.

7. Method 2: Reducing balance

A fixed percentage of the **net book value** (not cost) is charged each year. The charge is high in early years and falls each year.

Formula:

$$\text{Depreciation} = \text{NBV at start of year} \times \text{percentage}$$

Worked example 10.2

Equipment costs \$10,000. Depreciation is 20% per year, reducing balance.

Year	NBV at start	Depreciation (20%)	Accumulated depreciation	NBV at end
1	10,000	2,000	2,000	8,000
2	8,000	1,600	3,600	6,400
3	6,400	1,280	4,880	5,120
4	5,120	1,024	5,904	4,096

Best for: assets which lose more value in the early years and need more repairs later, for example motor vehicles. High depreciation early plus low repairs balances with low depreciation later plus high repairs.

8. Method 3: Revaluation

Used for many small, low cost items such as loose tools. It is too slow to depreciate each item.

Formula:

$$\text{Depreciation} = \text{Opening value} + \text{Purchases in the year} - \text{Closing value (valuation)}$$

Worked example 10.3

	\$
Loose tools at start of year	1,200
Add: Tools bought during year	500
	1,700
Less: Loose tools valued at end of year	1,100
Depreciation for the year	600

9. Buying part way through the year

Businesses choose a policy and use it consistently:

- **Full year** charge in the year of purchase and none in the year of sale, **or**
- **Monthly (time) basis:** charge only for the months owned.

Example: a machine costing \$12,000 is bought on 1 April. Depreciation is 10% per year on cost. Year end is 31 December.

Full year charge = $12,000 \times 10\% = \$1,200$. Monthly basis for 9 months = $1,200 \times 9/12 = \$900$.

Always read the question to see which policy to use.

10. Recording depreciation in the ledger

The asset stays at **cost** in its own account. Depreciation builds up in a separate account.

Entry	Debit	Credit
Depreciation for the year	Statement of Profit or Loss (depreciation expense)	Provision for depreciation account

Worked example 10.4 (using example 10.1)

Machinery account

Dr		\$	Cr		\$
Year 1 Jan 1	Bank	20,000	Year 1 Dec 31	Balance c/d	20,000

Provision for depreciation of machinery account

Dr		\$	Cr		\$
Year 1 Dec 31	Balance c/d	4,500	Year 1 Dec 31	Income statement	4,500
Year 2 Dec 31	Balance c/d	9,000	Year 2 Jan 1	Balance b/d	4,500
			Year 2 Dec 31	Income statement	4,500
		9,000			9,000
			Year 3 Jan 1	Balance b/d	9,000

Statement of Financial Position (extract) at end of Year 2

Non-current assets	Cost \$	Accumulated depreciation \$	NBV \$
Machinery	20,000	9,000	11,000

11. Disposal of a non-current asset

When an asset is sold, open a **Disposal account**. It works out the profit or loss on sale.

Step	Debit	Credit	Amount
1. Transfer the cost	Disposal	Asset account	Original cost
2. Transfer the depreciation	Provision for depreciation	Disposal	Accumulated depreciation to date
3. Record money received	Bank (or Cash)	Disposal	Sale proceeds
4. Balance the account			Profit or loss

- If the **credit** side is bigger, the difference is a **profit** on disposal (income).
- If the **debit** side is bigger, the difference is a **loss** on disposal (expense).

Quick check: Profit or loss = Sale proceeds - NBV at date of sale.

Worked example 10.5

A van cost \$30,000. Accumulated depreciation to the date of sale is \$18,000. It is sold for \$10,000 cash paid into the bank.

NBV = 30,000 - 18,000 = 12,000. Proceeds 10,000. **Loss = 2,000.**

Disposal of van account

Dr		\$	Cr		\$
	Van (cost)	30,000		Provision for depreciation	18,000
				Bank	10,000
				Income statement (loss on disposal)	2,000
		30,000			30,000

If the van had been sold for \$13,000, the profit would be \$1,000 and it would appear on the **debit** side as "Income statement (profit on disposal)".

12. Where does it all go?

Item	Statement of Profit or Loss	Statement of Financial Position
Depreciation for the year	Expense	
Accumulated depreciation		Deducted from cost under non-current assets
Profit on disposal	Other income (added to gross profit)	
Loss on disposal	Expense	
Cost of asset		Non-current assets

13. Common mistakes

- Using **cost** instead of **NBV** for reducing balance.
- Forgetting to deduct residual value in straight line.
- Recording depreciation in the asset account instead of the provision account.
- Treating repairs as capital expenditure. Repairs keep an asset working; they do not improve it.
- Mixing up the sides of the disposal account. Profit balances on the **debit** side.

14. Quick summary

- Non-current assets are used, not sold, and last more than a year.
- Depreciation spreads cost over useful life (accruals) and avoids overstating assets (prudence).
- Straight line: same amount yearly. Reducing balance: percentage of NBV. Revaluation: opening + purchases - closing.
- Disposal: proceeds minus NBV equals profit or loss.

15. Check yourself

1. A computer costs \$3,000 plus \$200 installation. It will last 4 years with no residual value. Find the straight line depreciation.
2. A car costs \$16,000. Depreciation is 25% reducing balance. Find the NBV after 2 years.
3. A machine with NBV \$4,000 is sold for \$4,600. Profit or loss?

Answers: see [model-questions/Answer_Key.md](#) , Chapter 10 section.